

→ Rest and motion are relative.

→ No absolute rest. No Absolute motion.

→ Both are defined w.r.t a "Frame of Reference".

★ Frame of Reference = An observer coordinate system + a clock, attached to rigid body.

★ Dimensions of Motion :-

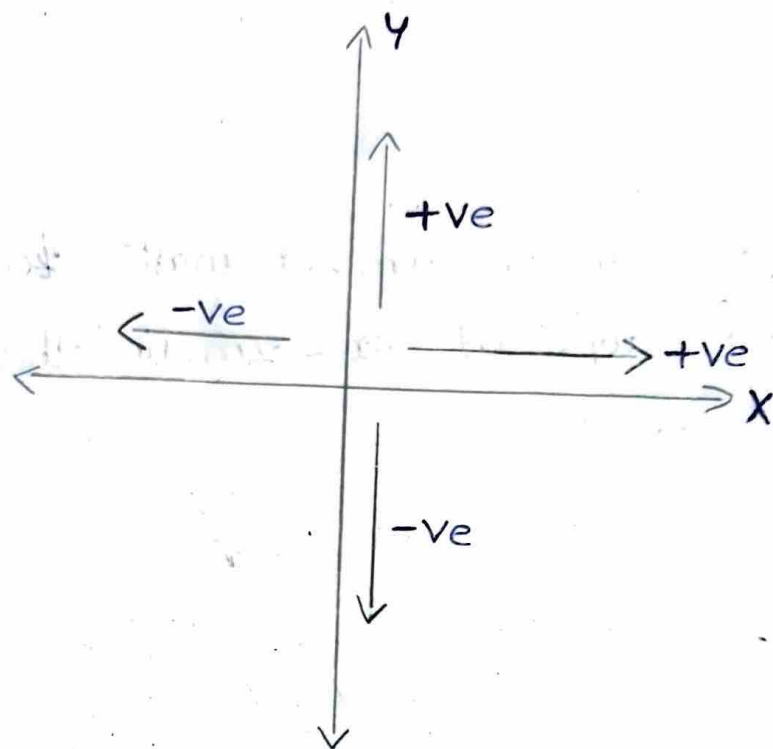
1D motion → 1 co-ordinate changes with time.

2D motion → 2 co-ordinates changes with time.

3D motion → 3 co-ordinates changes with time.

Sign Convention :-

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★ Distance and Displacement.

Distance :- i) Total length of the path travel.

ii) Scalar $\rightarrow$ No direction

iii) It is always greater than (or) equal to zero.

iv) Can only increase (or) stay the same.

Displacement :- i) It is the shortest path between initial and final positions.

ii) It is change in position, $\Delta x = \text{change in position}$

$$\Delta x = x_{\text{final}} - x_{\text{initial}}$$

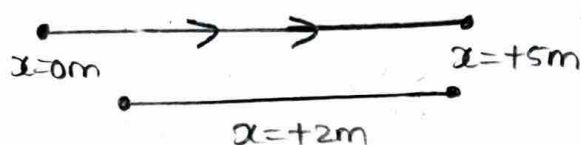
iii) Vector $\rightarrow$ Sign carries direction in 1D.

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iv) It can be positive, zero (or) negative.

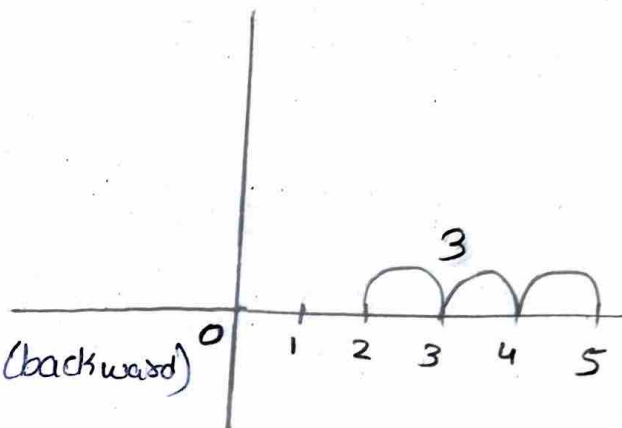
Ex: 1 - A particle starts at $x = 0\text{m}$. It walks to $x = +5\text{m}$ then turn back and stops at $x = 2\text{m}$. Find the distance & displacement.

Sol:



$$\text{Distance} = 5\text{m (forward)} + 3\text{m (backward)}$$

$$= 8\text{m}$$

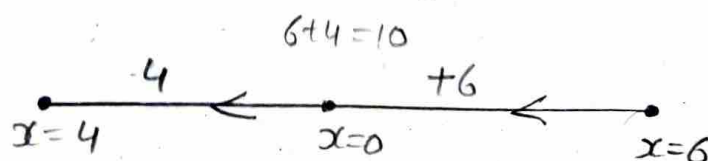


$$\text{Displacement} = \Delta x = x_f - x_i$$

$$= (+2) - (0) = +2\text{m}$$

$$\Delta x = +2\text{m}$$

Ex: 2. A particle moves from $x = +6\text{m}$ to $x = -4\text{m}$ find the distance and displacement.



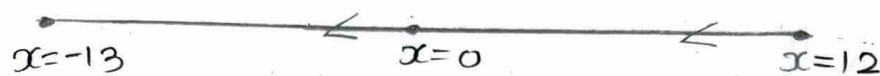
$$\text{Distance} = 6+4 = 10\text{m}$$

$$\text{Displacement} = \Delta x = x_f - x_i \Rightarrow -4 - (+6) = -10$$

$$\Delta x = -10\text{m}$$

Ex: 3 A particle moves from $x = +12\text{m}$ to $x = -13\text{m}$
find distance and displacement.

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$$\begin{aligned}\text{Distance} &= 12 + 13 \\ &= 25\text{m}\end{aligned}$$

$$\begin{aligned}\text{Displacement} &= \Delta x = x_f - x_i \\ &= -13 - (+12) \\ &\Rightarrow -13 - 12 = -25\end{aligned}$$

$$\boxed{\Delta x = -25}$$

★ Key Results:

→ Distance $\geq$ |Displacement| → Always True

→ Equal only when straight line/no reversal.

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*) Distance = 400 m

Time = 80 s

Displacement = 0 m

$$\text{Average speed} = \frac{400}{80} = 5 \text{ m/s}$$

$$\text{Average velocity} = \frac{0}{80} = 0 \text{ m/s.}$$

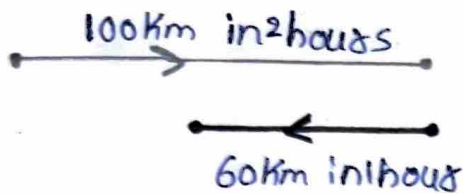
*) Average speed & Average velocity

$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total time}} = \frac{d}{t}$$

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Total time}} = \frac{\Delta x}{t}$$

* Ex:1 - A car goes 100 km east in 2 hrs, then 60 km west in 1 hour. Find the average speed & velocity. Harshith

Sol:-



$$\text{Distance} = 100 + 60 = 160 \text{ km}$$

$$\text{Time} = 3 \text{ hrs}$$

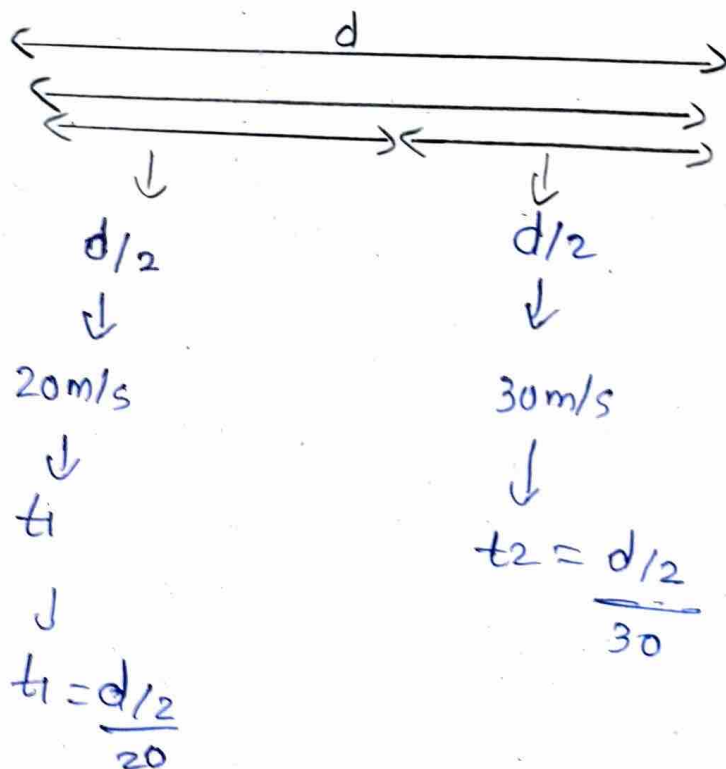
$$\text{Avg speed} = \frac{160}{3}$$

$$\text{Disp} = 40 \text{ km}$$

$$\text{Avg velocity} = \frac{40}{3}$$

* A car covers the first half of distance at 20 m/s, the second at 30 m/s. Find the average speed.

Sol:



$$t_1 = \frac{d}{40}$$

$$t_2 = \frac{d}{60}$$

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$$t_1 + t_2 = \frac{d}{40} + \frac{d}{60}$$

$$= \frac{100d}{2400} = \frac{1}{24} \Rightarrow t = \frac{d}{24}$$

$$\text{Average speed} = \frac{d}{\frac{d}{24}} \Rightarrow 24 \text{ m/s}$$

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10) A car goes at 60 km per hour for 2 hours. then it goes at 40 km per hour for 2 hours. Find the average speed.

Sol:

$$\text{Avg speed} = \frac{\text{Total Distance}}{\text{Total Time}} \Rightarrow \frac{d_1 + d_2}{d_1 + d_2}$$

$$s = \frac{d}{t}$$

$$d_1 = 60 \times 2$$

$$d_2 = 40 \times 2$$

$$d_1 = 120$$

$$d_2 = 80$$

$$= \frac{120 + 80}{4}$$

$$= \frac{200}{4} = 50 \text{ km/hr}$$

→ Equal distances at speed v_1 and v_2 :

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$$\text{Avg speed} = \frac{2v_1 + v_2}{v_1 + v_2} \text{ (H.M.)}$$

→ Equal times at speed v_1 and v_2 :

$$\text{Avg speed} = \frac{v_1 + v_2}{2} \text{ (simple Average)}$$

★ Instantaneous velocity!

start with average velocity over Δt ,

$$v_{\text{Avg}} = \frac{\Delta x}{\Delta t}$$

make the interval smaller and smaller:

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Instantaneous velocity = derivative of position w.r.t time.

Eg: Position, $x(t) = 3t^2 + 2t$ (m, s)

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$$t=0$$

$$x(0) = 3(0)^2 + 2(0)$$

$$= 0$$

$$t=1$$

$$x(1) = 3(1)^2 + 2(1)$$

$$= 3 + 2$$

$$= 5 \text{ m}$$

$$v = \frac{dx}{dt} = 3(2t) + 2$$

$$v = 6t + 2$$

$$v \text{ At } t=0 \Rightarrow 6(0) + 2 = 2$$

$$t=1 \Rightarrow 6 + 2 = 8$$

$$t=5 \Rightarrow 6(5) + 2 = 32$$

→ Velocity is different at every instant. This is called non-uniform motion.

$$\text{Instantaneous speed} = |v| = \left| \frac{dx}{dt} \right|$$

* Uniform and non-uniform motion:

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→ Uniform motion:- i) velocity is constant.

ii) Average velocity = Instantaneous velocity.

iii) Acceleration = 0

iv) Eg:- Cruise Control on Highway.

→ Non-uniform motion:- i) Velocity changes with time.

ii) Average velocity $\neq$ Instantaneous velocity.

iii) Acceleration $\neq 0$

iv) Eg:- Driving in city traffics.

$$x(t) = 5t^3 + 2t \rightarrow \text{find}$$

v at $t=0, 1,$
5sec

★ Momentarily at Rest!

Eg: $x = t^3 - 6t^2 + 9t$

find $v(t)$ and the instant (s) where the ^{when} the particle is momentarily at Rest.

$$\frac{dx}{dt} = 3t^2 - 12t + 9$$

$$v(t) = 3t^2 - 12t + 9$$

$$3t^2 - 12t + 9 = 0$$

$$t^2 - 4t + 3 = 0$$

$$t^2 - 3t - t + 3 = 0$$

$$t(t-3) - 1(t-3) = 0$$

$$(t-1)(t-3) = 0$$

$$t = 1s \text{ (or) } t = 3s$$

$$\rightarrow v = 3t^2 - 12t + 9$$

$$\stackrel{t=1}{=} 3(1)^2 - 12(1) + 9$$

$$v = 3 - 12 + 9 = 0$$

$$t=3$$

$$\rightarrow v = 3(3)^2 - 12(3) + 9$$

$$= 3(9) - 36 + 9$$

$$= 27 - 36 + 9 = 0$$

$$=$$

momentarily at Rest $\Rightarrow$ velocity = 0, at certain point.

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$\rightarrow$ It doesn't mean it stops, it means particle is turning around.

★ Acceleration (Accⁿ) :-

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Accⁿ = rate of change of velocity.

= derivate of velocity

= second derivate of position.

Eg: $x = 3t^2 + 2t$. find a.

$$\frac{dx}{dt} = \frac{d}{dt}(3t^2 + 2t) \Rightarrow 6t + 2$$

$$\frac{dx}{dt} = 6t + 2$$

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

*) Acceleration is constant (6m/s^2 does not depend on t)

*) This is uniformly accelerated motion.

* Misconception - 1 :-

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Fast = high acceleration

↓
wrong!

- A car at 200 km/h with constant speed has "zero" accⁿ. But a car going from 0 to 30 km/h has non-zero accⁿ.
- Accⁿ is about change in velocity, not about mean fast.

* Misconception - 2 :-

Negative accⁿ = slowing down

↓
wrong

- Negative accⁿ means, accⁿ points in the negative x direction.
- Speeding up (or) slowing down depends up on accⁿ and velocity, have the same (or) opposite signs

$v > 0, a > 0 \rightarrow$ speeding up (both +ve)

$v > 0, a < 0 \rightarrow$ slowing down (opposite)

$v < 0, a < 0 \rightarrow$ speeding up (both -ve)

$v < 0, a > 0 \Rightarrow$ slowing down (opposite)

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★) Equations of motion !

→ Important :- Acceleration must be constant.

u = initial velocity

v = final velocity

a = accⁿ

s = disp

t = time

$$1) v = u + at$$

$$2) s = ut + \frac{1}{2}at^2$$

$$3) v^2 = u^2 + 2as$$

→ If acceleration changes with time or position, these equations do not work.

→ Use when "constant accⁿ", "uniform accⁿ"

"free fall", "constant force on constant mass".

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- ★) A car accelerates uniformly from rest at 2m/s^2 for 8s. Find the final velocity and distance covered.

Sol: $t = 8 \text{ sec}$

$a = 2\text{m/s}^2$

$V = u + at$

$V = 0 + 2 \times 8 = 16$

$V = 16\text{m/s}$

$S = ut + \frac{1}{2} at^2$

$S = 0 + \frac{1}{2} \times 2 \times 8 \times 8$

$S = 8 \times 8 = 64$

$S = 64\text{m}$

- ★ A train moving at 20m/s brakes with deceleration 2m/s^2 . How far does it travel before stopping.

Sol:

$V^2 = u^2 + 2as$

$0 = (20)^2 + 2(-2)s$

$(20)^2 + 2(-2)s = 0$

$400 + (-4)s = 0$

$400 - 4s = 0$

$-4s = -400$

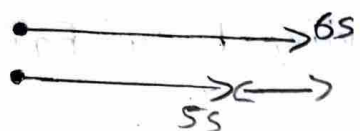
$s = \frac{+400}{+4} = 100$

$S = 100\text{m}$

★ Displacement in n^{th} second :-

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$$S_n = (\text{dist in } n \text{ sec's}) - (\text{dist in } (n-1) \text{ sec's})$$


$$= \left(un + \frac{1}{2} an^2 \right) - \left(u(n-1) + \frac{1}{2} a(n-1)^2 \right)$$
$$= (un - u(n-1)) + \frac{1}{2} a(n^2 - (n-1)^2)$$

$$S_n = u + \frac{1}{2} a (2n-1)$$

Q) A body is at rest initially and starts moving with uniform acca with 2m/sec^2 . Find distance covered in 5 sec and displacement in 5th second.

Sol: $u = 0\text{m/s}$

$t = 5\text{m/s}$

$a = 2\text{m/s}^2$

$$S_n = u + \frac{1}{2} a (2n-1)$$

$$S_5 = 0 + \frac{1}{2} \times 2 \times (2 \times 5 - 1)$$

$$S_5 = 2 \times 5 - 1$$

$$S_5 = 10 - 1$$

$$S_5 = 9\text{m}$$

$$S = ut + \frac{1}{2} at^2$$

$$= 0 + \frac{1}{2} \times 2 \times 25$$

$$S = 25\text{m}$$

* Motion under gravity

→ Near earth's surface every freely moving body has the same down ward acceleration.

$$g = 9.8 \text{ m/s}^2 \text{ (use } 10 \text{ m/s}^2 \text{ for quick answers).}$$

→ This is independent of the objects mass. It is uniformly accelerated motion, the same three equations apply directly.

* The sign convention steps:-

Step 1:- Choose a positive direction.

We will use up = +ve (Always)

Step 2:- Everything follows that one choice.

$$\Rightarrow a = g = -9.8 \text{ m/s}^2$$

$$\Rightarrow \text{acc}^n \text{ in down} = -ve$$

$$\Rightarrow v \text{ in up} = +ve$$

$$\Rightarrow v \text{ in down} = -ve$$

$$\Rightarrow \text{acc}^n \text{ in up} = +ve$$

step 3:- Plug in to the three equations.

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$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

step 4:- Never change the convention mid problem.

Ex:- A ball is dropped from a 45 m tower. Find the time to reach the ground and the speed on landing.

sol:

up = positive (+ve)

$$u = 0 \text{ m/s}$$

$$a = -g = -10 \text{ m/s}^2$$

$$s = -h = -45 \text{ m}$$

$$s = ut + \frac{1}{2}at^2$$

$$-45 = 0(t) + \frac{1}{2}(-10)(t)^2$$

$$-45 = -5t^2$$

$$9 = t^2 \Rightarrow t^2 = 9$$

$$t = 3 \text{ s}$$

$$v = u + at$$

$$v = 0 + (10)(3)$$

$$v = (10)(3)$$

$$v = -30 \text{ m/s}$$

Ex:- A ball is thrown down at 5 m/s ~~at~~ from the same 45 m tower. find the time to reach the ground. Harshith

up = +ve

down = -ve

$$u = -5 \text{ m/s}$$

$$a = -g = -10 \text{ m/s}^2$$

$$s = -45 \text{ m}$$

$$s = ut + \frac{1}{2} at^2$$

$$-45 = -5(t) + \frac{1}{2}(-10)(t)^2$$

$$9 = t + t^2$$

$$t^2 + t - 9 = 0$$

$$\therefore \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1 - 4(1)(-9)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{37}}{2}$$

$$t = 2.55 \text{ s}$$

$$\frac{-1 - \sqrt{37}}{2} \quad \text{and} \quad \frac{-1 + \sqrt{37}}{2} \Rightarrow \frac{-1 + \sqrt{37}}{2}$$

$$t = 2.5$$

Ex:- A ball is thrown down at 5m/s ~~at~~ from the same 45m tower. find the time to reach the ground. Harshith

up = +ve

down = -ve

$$u = -5 \text{ m/s}$$

$$a = -g = -10 \text{ m/s}^2$$

$$s = -45 \text{ m}$$

$$s = ut + \frac{1}{2} at^2$$

$$-45 = -5(t) + \frac{1}{2}(-10)(t)^2$$

$$9 = t + t^2$$

$$t^2 + t - 9 = 0$$

$$\therefore \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1 - 4(1)(-9)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{37}}{2}$$

$$t = 2.55 \text{ s}$$

$$\frac{-1 - \sqrt{37}}{2} \quad \text{and} \quad \frac{-1 + \sqrt{37}}{2} \Rightarrow \frac{-1 + 6.1}{2}$$

$$t = 2.55 \text{ s}$$

"Dropped" $\Rightarrow u=0$

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→ Thrown down implies $u \neq 0$, u is in down direction. (-ve).

→ Thrown up implies $u \neq 0$, u is in up direction (+ve).

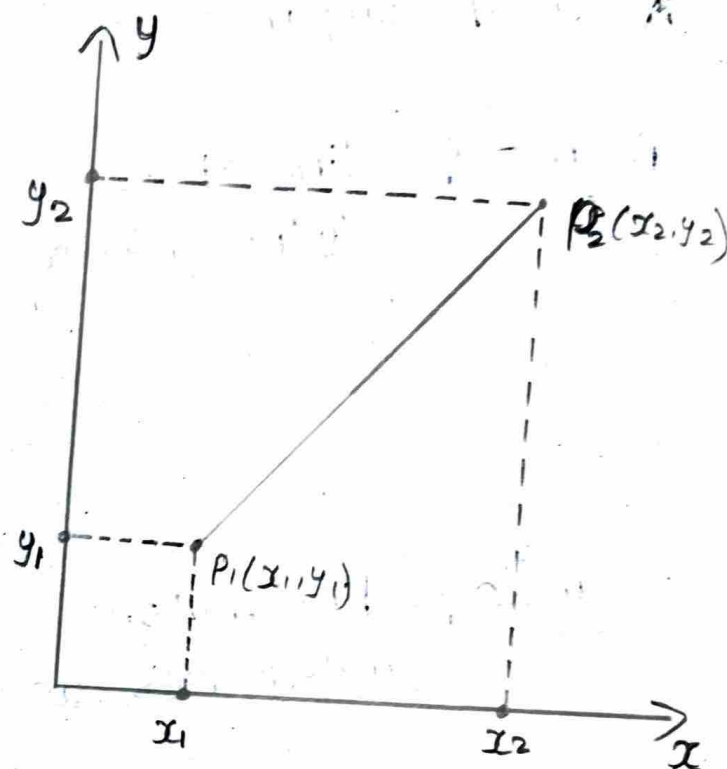
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★ Slope :-

Take any two points on a line, straight line:

$P_1(x_1, y_1)$ and $P_2(x_2, y_2)$

$$\text{slope} = m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$



Eg: 1 $P_1(2, 4)$ $P_2(4, 6)$

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 4}{4 - 2} = \frac{2}{2} = 1$$

Eg: 2 $P_1(0, 10)$ $P_2(5, 0)$

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 10}{5 - 0} = \frac{-10}{5} = -2$$

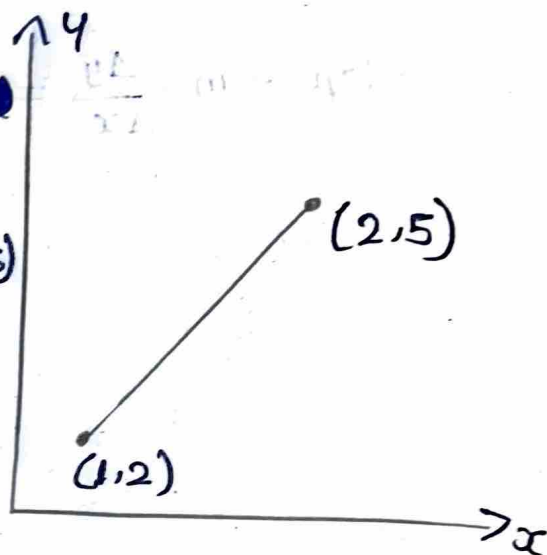
Eg: 3 $P_1(1,4)$ $P_2(6,4)$
 $x_1 y_1$ $x_2 y_2$

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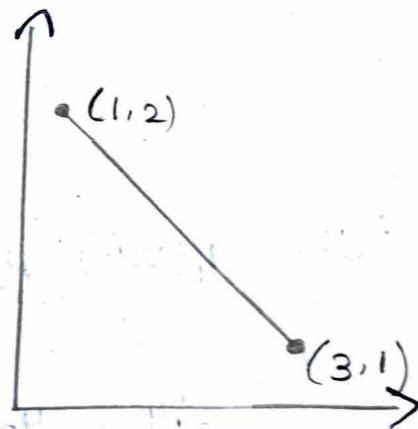
$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 4}{6 - 1} = \frac{0}{5} = 0$$

★ Interpreting m :

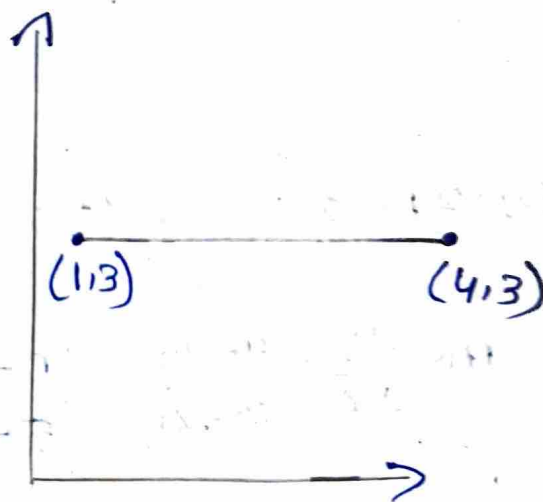
- 1) $m > 0 \Rightarrow$ line rises
 (y increases as x increases)



- 2) $m < 0 \Rightarrow$ line falls.
 (y decreases as x increases)

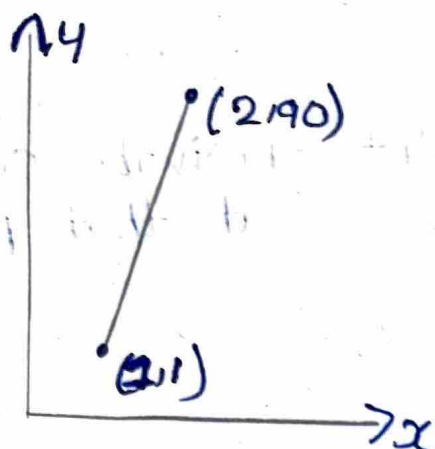


- 3) $m = 0 \Rightarrow$ line is horizontal
 (y is constant)

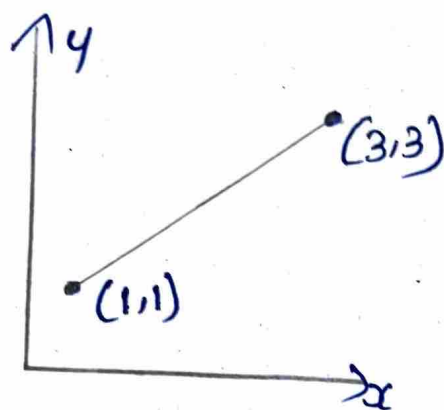


4) If $|m|$ is very large $\Rightarrow$ steep line

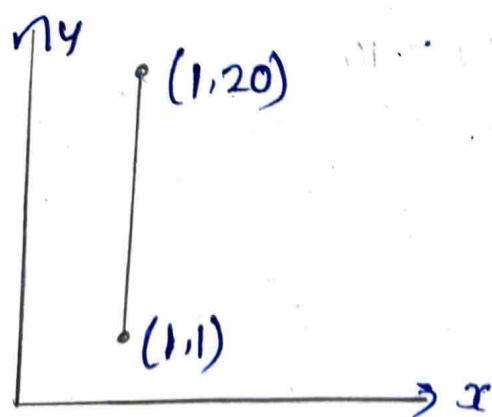
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5) If $|m|$ is very small $\Rightarrow$ gentle / shallow line.



6) If m is undefined $\Rightarrow$ line is vertical ($\Delta x = 0$)



*)

slope = derivative

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slope of a curve at a point = Derivative of the curve at that point.

→ Eg: $y = x^2$

find slope at $x = 1$

$$m = \frac{dy}{dx}$$

$$y = x^2$$

$$\frac{dy}{dx} = \frac{d}{dx} x^2 = 2x$$

$$\text{slope at } x=1 = 2x \Rightarrow 2(1) = 2$$

$$x = -1 = 2(-1) = -2$$

$$x = 5 = 2(5) = 10$$

→ Eg: $y = 4x^3$

find slope at $x = 2, 3$

$$y = 4x^3$$

$$\frac{dy}{dx} = \frac{d}{dx} 4x^3 = 12x^2$$

$$x = 2, 3 \Rightarrow 12(2)^2 = 12 \times 4 = 48$$

$$= 12(3)^2 = 12 \times 9 = 108$$

* A ball is thrown straight up at $u = 20 \text{ m/s}$, take g as 10 m/s^2 .
~~What is~~ find time taken for ball to reach the top.
 And maximum height.

Sol: $V = 0$

$$u = 20 \text{ m/s}$$

$$h_{\text{max}} = ?$$

time taken to reach the top = ?

$$V = u + at$$

$$0 = 20 + (-10)t$$

$$0 = 20 - 10t$$

$$-20 = -10t$$

$$-10t = -20$$

$$t = \frac{-20}{-10} = 2$$

$$\boxed{t = 2 \text{ sec}}$$

$$V^2 = u^2 + 2as$$

$$V^2 - u^2 = 2as$$

$$-(20)^2 = 2(-10)s$$

$$-400 = -20s$$

$$-20s = -400$$

$$s = \frac{-400}{-20} = 20$$

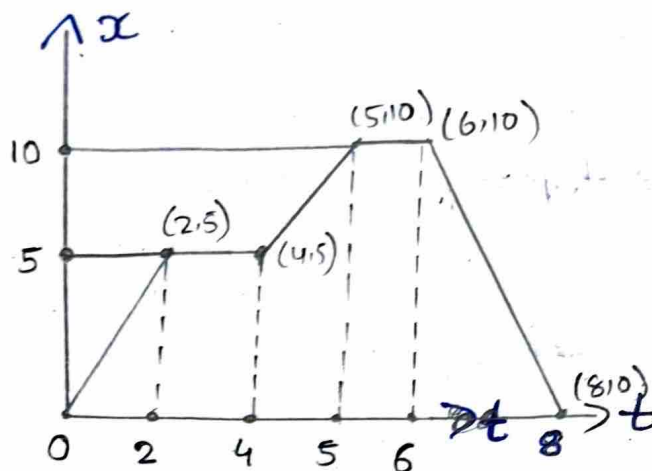
$$\boxed{s = 20 \text{ m}}$$

* Position $x-t$ $\xrightarrow[\text{slope}]{\text{Differentiate}}$ velocity

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velocity $v-t$ $\xrightarrow[\text{slope}]{\text{Differentiate}}$ acceleration $a-t$

Slope = Differentiation

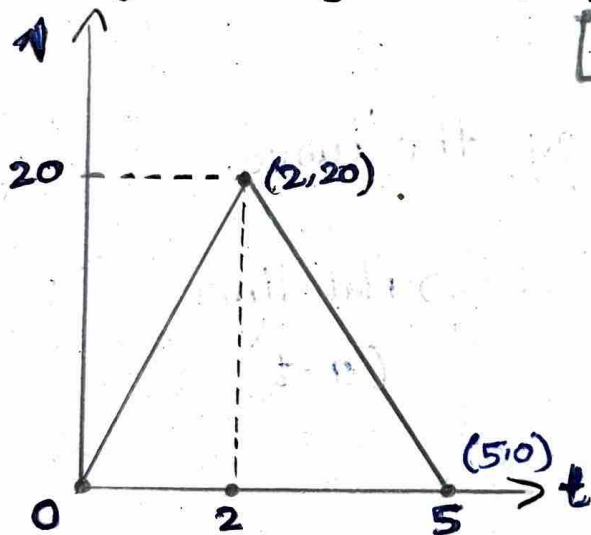


- 1) Horizontal line $\Rightarrow$ slope "0". $\Rightarrow$ At rest.
- 2) straight slanted line $\Rightarrow$ uniform velocity.
- 3) steeper line implies faster or more speed.
- 4) upward slope (+) $\Rightarrow$ moving in positive direction.
- 5) Downward slope (-) $\Rightarrow$ moving in negative x direction.
- 6) curved line implies ~~the~~ change in velocity.

* Misconception:-

- 1) An $x-t$ graph is not a picture of the path.
- 2) A rising $x-t$ line does not mean the object moves "upward." It means the position coordinate increases with time, nothing about actual direction in space.
- 3) Ex: A flat road can still have a curved $x-t$ graph

* Reading velocity time graphs:-



Slope of $v-t$ graph = Acc^n .

from $t=0$ to $2s$

$$m = \frac{(20-0)}{2-0} = 10$$

$$a = 10 \text{ m/s}^2$$

from $t=2$ to $5s$

$$m = \frac{(0-20)}{5-2} = \frac{-20}{3} = -6.66$$

$$a = -6.66 \text{ m/s}^2$$

- 1) Rising line $\Rightarrow$ Positive acceleration.
- 2) Falling line $\Rightarrow$ negative acceleration.
- 3) Horizontal line $\Rightarrow$ "0" acceleration (uniform velocity)
- 4) steeper line implies larger acceleration.
- 5) Straight line $\Rightarrow$ uniform acceleration.

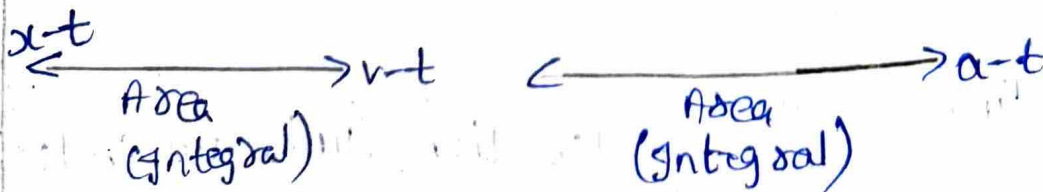
★ The step Ladder :-

Motion Type	$x-t$ graph	$v-t$ graph	$a-t$ graph
At rest	Horizontal line	on the axis	on the axis
uniform velocity	straight slope	Horizontal line	on the axis
uniform Acc ⁿ	PARABOLA	straight slope	Horizontal line

*) Diffⁿ lowers the degree of the curve.

Parabola $\rightarrow$ straight line $\rightarrow$ Flat line
 $(x-t)$ $(v-t)$ $(a-t)$

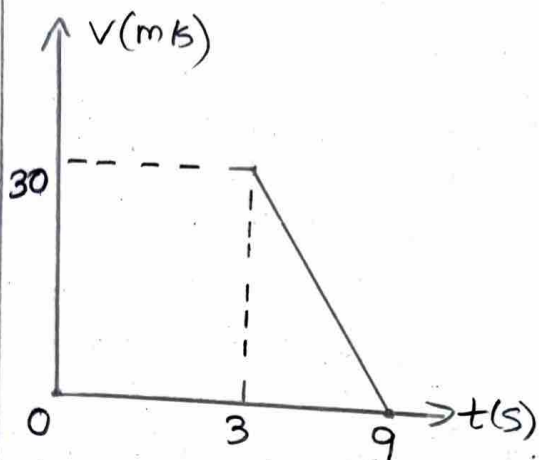
$\text{Area} = \text{Integral}$



*) Area under $v-t$ graph (B/w two points) is equal to displacement in the time interval.

* Area under a graph (b/w two points) = change in velocity.
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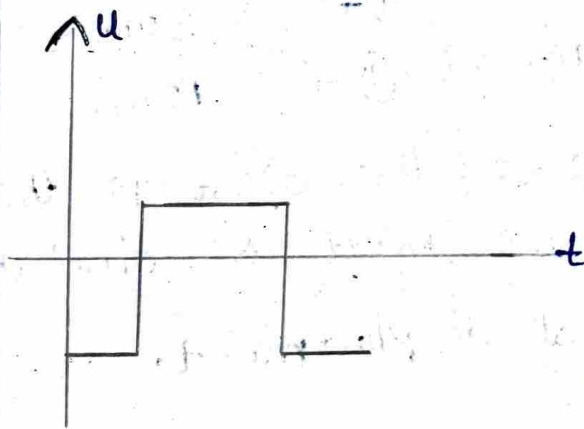
* Example:-



$$\text{Area} = \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times 9 \times 30 = 135$$

* The sign of the area:-



* above the x -axis the velocity is positive. The area gives +ve displacement.

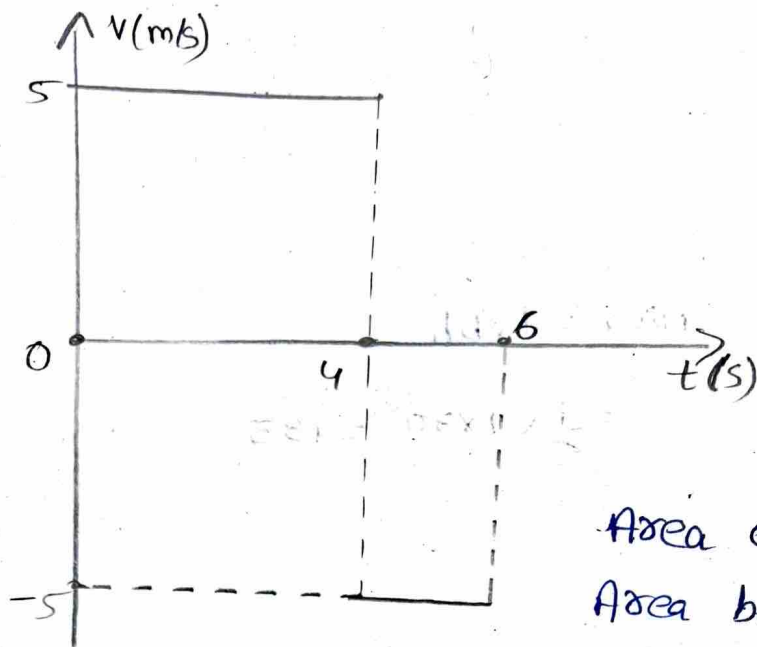
* below the x -axis the velocity is negative. the area gives -ve displacement.

Eg: distance and displacement from the graph.

displacement = area above axis - area under axis.

distance = area above axis + area below axis

eg: A vt graph is +5m/s for the first 4 sec then -5m/s for next 2 sec. find the distance and displacement



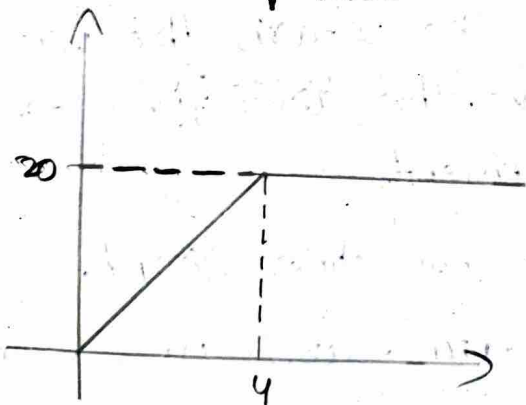
Area above x axis = $4 \times 5 = 20\text{m}$ ①

Area below x axis = $2 \times 5 = 10\text{m}$ ②

Distance = ① + ② = 30m

Displacement = ① - ② = 10m

e) A vt graph from 0 to 20m/s for the first 4s then it says constant at 20m/s for next 6s find the accⁿ in phase A and total displacement.



$a = \text{slope}$

$= \frac{20-0}{4-0}$

5m/s^2

$a = 5\text{m/s}^2$

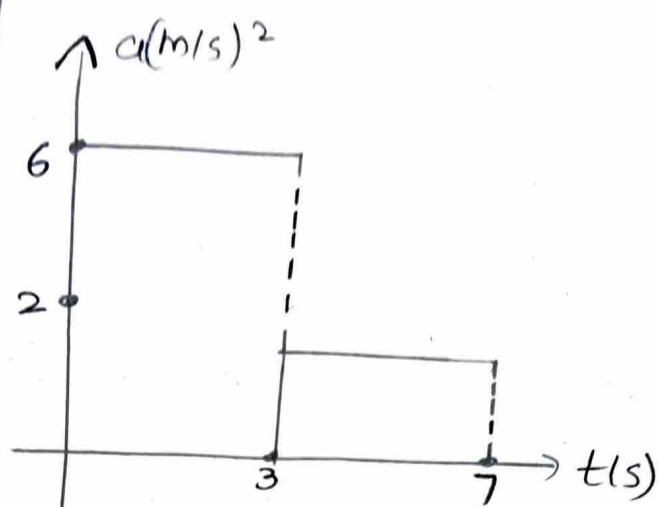
Disp = Area of Δ + Area of rectangle =

$\frac{1}{2} \times b \times h +$

$= \frac{1}{2} \times 20 \times 4 + 20 \times 6$

$= 40 + 120 = 160\text{m}$

disp = 160m



Initial velocity = 5 m/s

$$\text{Area of A} = 3 \times 6 = 18$$

$$\text{Area of B} = 4 \times 2 = 8$$

$$\text{Total area} = 26$$

$$\Delta v = 20 \text{ m/s}$$

$$\Delta v = v_f - v$$

$$26 = v_f - 5 \Rightarrow v_f - 5 = 26 \Rightarrow v_f = 26 + 5 = 31$$

$$v_f = 31 \text{ m/s}$$

$$\int_a^b f(x) dx = \left[f(x) \right]_a^b = F(b) - F(a)$$

a = lower limit = Initial value

b = upper limit = final value.

* Three working formula's

Harshith

1) $a = \frac{dv}{dt}$

2) $v = \frac{dx}{dt}$

3) $a = v \times \frac{dv}{dx}$

$\frac{dv}{dt} \times \frac{dx}{dt} \rightarrow (v)$

$a = v \times \frac{dv}{dx}$

* If 'a' depends on

1) Time : $a = f(t)$

2) Position : $a = f(x)$

3) Velocity : $a = f(v)$

use

~~a = f(t)~~

$a = \frac{dv}{dt}$

$a = \frac{v dv}{dx}$

$a = \frac{dv}{dt} \text{ (or) } a = \frac{v dv}{dx}$

Q1) $a=6t$, Particle starts from rest at the origin. Find v and displacement at $t=2\text{sec}$. Harshith

$$a=6t \Rightarrow a = \frac{dv}{dt}$$

$$\text{at } x=0, v=0$$

$$\boxed{v=3t^2} \rightarrow \therefore \frac{dv}{dt} = 6t \Rightarrow \int dv = \int 6t dt$$

$$v = 6 \int t dt$$

$$v = \frac{6t^2}{2} + C$$

$$\boxed{v=3t^2+C}$$

$$\boxed{v=3t^2}$$

$$\boxed{0=0+C}$$

$$\frac{dx}{dt} = 3t^2$$

$$dx = 3t^2 \times dt$$

$$\int dx = \int 3t^2 \times \int dt$$

$$x = \frac{3t^3}{3} + C, \quad \boxed{x=t^3}$$

$$\boxed{x=t^3}$$

$$\boxed{x=8\text{m}}$$

$$\boxed{0=0+C}$$

Q2) $a=4x$, Particle is at $x=1\text{m}$ with $v=2\text{m/s}$ find v where $x=3\text{m}$.

$$a=4x, \quad x=1\text{m}, \quad \text{with } v=2\text{m/s}$$

$$a = \frac{v dv}{dx}$$

$$\frac{v dv}{dx} = 4x$$

$$v dv = 4x dx$$

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$$\int_1^v v dv = 4 \int_1^3 x dx$$

$$\left[\frac{v^2}{2} \right]_1^v = 4 \left[\frac{x^2}{2} \right]_1^3$$

$$\frac{v^2}{2} - \frac{1^2}{2} = 4 \left(\frac{3^2}{2} - \frac{1^2}{2} \right)$$

$$\frac{v^2}{2} - 2 = 4 \left(\frac{9-1}{2} \right)$$

$$\frac{v^2}{2} - 2 = 2(8) = 16$$

$$\frac{v^2}{2} = 16 + 2 = 18$$

$$v^2 = 36 \Rightarrow v = \sqrt{36} \Rightarrow v = \sqrt{(6)^2}$$

$$v = \pm 6$$

$$v = 6 \text{ m/s}$$

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*)

The relation between time t and distance x for a moving body is given as $t = mx^2 + nx$, where m and n are constants. The retardation of the motion is: (when v stands for velocity).

sol:

$$t = mx^2 + nx$$

differentiate with resp to time

$$\frac{dt}{dt} = \frac{m d(x^2)}{dt} + n \frac{dx}{dt}$$

$$1 = m \frac{d(x^2)}{dx} \times \frac{dx}{dt} + nv \quad \left(\frac{dx}{dt} = v \right)$$

$$1 = 2xm v + nv$$

$$1 = v(2xm + n)$$

$$v = \frac{1}{2xm + n} \Rightarrow (2xm + n)^{-1}$$

$$\frac{dv}{dt} = \frac{d}{dt} (2xm + n)^{-1}$$

$$a = -1 \times (2xm + n)^{-2} \times 2m \frac{dx}{dt}$$

$$a = -1 \times v^3 \times 2mv$$

$$a = -2mv^3$$

$$\boxed{a = 2mv^3}$$

Formula 1: $v = dx/dt$

Harshith

1) The position of a particle is

$$x = 4t^2 - 3t + 2$$

$$v = \frac{dx}{dt} = \frac{d}{dt}(4t^2 - 3t + 2)$$

$$v = 8t - 3$$

$$a = \frac{dv}{dt} = \frac{d(8t - 3)}{dt} = 8$$

$$\boxed{a = 8}$$

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*) Relative velocity

$$\text{Relative velocity} = V_{AB} = V_A - V_B$$

$$*) V_A \text{ with respect to } B = V_A - V_B$$

*) Both V_A and V_B are measured from the same fixed ground frame.

$$*) V_{AB} = V_A \text{ as seen by } B \Rightarrow V_A - V_B$$

$$*) V_{BA} = V_B \text{ as seen by } A \Rightarrow V_B - V_A$$

⇒ Ex: two cars A & B are moving in the same direction
(both moving right) $V_A = 60$ $V_B = 40$

sol: $V_{AB} = V_A - V_B$

$$V_{AB} = 60 - 40$$

$$V_{AB} = 20$$

car A moves away from car B at 20 m/s.

⇒ Ex: car A & car B are moving in opposite A = right,
B = left. $V_A = 60$ $V_B = -40$

$$V_{AB} = V_A - V_B$$

$$V_{AB} = 60 - (-40)$$

$$V_{AB} = 60 + 40 = 100$$

$$V_{AB} = 100$$

Eg:- Car A

 $\xrightarrow{30 \text{ m/s}}$

Car B

 $\xrightarrow{20 \text{ m/s}}$

100m

method-I $\therefore$ Ground's FOR

$$x_A = 0 + v_A t \Rightarrow 0 + 30t$$

$$x_B = 100 + v_B t \Rightarrow 100 + 20t$$

$$x_A = x_B \Rightarrow 30t = 100 + 20t$$

$$\Rightarrow 10t = 100$$

$$\Rightarrow t = \frac{100}{10} = 10$$

$$\boxed{t = 10}$$

method-II

$$v_{AB} = v_A - v_B = +30 - (+20)$$

$$\boxed{v_{AB} = 10 \text{ m/s}}$$

$$t = \frac{d}{v_{AB}} = \frac{100}{10} = 10 \text{ s}$$

$$\boxed{t = 10 \text{ s}}$$

* Frame switch method :-

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step 1 :- Pick which object is the observer.

The observer is always at rest. In their own FOR by definition.

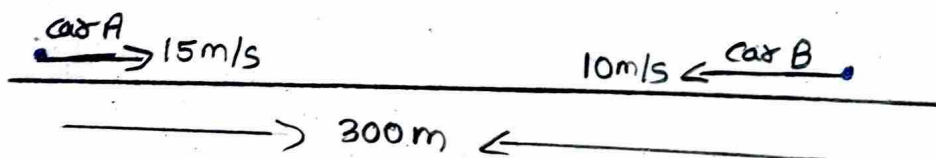
step 2 :- Find how the other object moves as seen by the observer.

$$V_{AB} = V_A - V_B.$$

step 3 :- Now solve as if only one object is moving, towards the fixed observer.

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Ex: 2 cars 300m apart move towards each other. Car A at 15m/s, car B at 10m/s find the time they meet, and the distance from A's starting point.



Sol:

B → observer, Car B's FOR (Frame of Reference)

$$V_{AB} = V_A - V_B$$

$$= 15 - (-10) \Rightarrow 25 \text{ m/s}$$

$$V_{AB} = 25 \text{ m/s}$$

$$t = \frac{d}{v_{AB}} = \frac{300}{25} = 12.5$$

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$$x_A = v_A t = 15 \times 12.5 = 180$$

$$x_A = 180 \text{ m}$$

Ex. 2:- A cyclist is moving at 8 m/s and he is ^{chasing} ~~chasing~~ a jogger at 10 m/s who is 20 m ahead in same direction when does the cyclist catch the jogger.

Sol: Cyclist cannot catch the jogger because the velocity of jogger is more than cyclist.

* motion In a plane

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Vector Recap

A vector A at angle θ to the x axis resolves into components:

$$A_x = A \cos \theta, \quad A_y = A \sin \theta$$

To recombine the components back into the full vector:

$$|A| = \sqrt{A_x^2 + A_y^2} \quad \theta = \tan^{-1}(A_y/A_x)$$

Rule: Always resolve a vector into perpendicular (x, y) components before doing kinematics. We solve along the axes, never along a slanted direction directly.

Example:- Resolve 10 m/s directed at 37° above the horizontal ($\sin 37^\circ \approx 0.6$, $\cos 37^\circ \approx 0.8$) find v_x and v_y .

Sol:

$$A_x \cos \theta = 10 \cos 37^\circ = 10 \times \frac{4}{5} = \frac{40}{5} = 8.0$$

$$A_y \sin \theta = 10 \sin 37^\circ = 10 \times \frac{3}{5} = \frac{30}{5} = 6.0$$

2) Vector in motion

Position : $\vec{r} = x\hat{i} + y\hat{j}$

velocity : $\vec{v} = d\vec{r}/dt = v_x\hat{i} + v_y\hat{j}$

Acceleration : $\vec{a} = d\vec{v}/dt = a_x\hat{i} + a_y\hat{j}$

Differentiation in 2D is componentwise. The x-part of $\vec{v}$ depends only on the x-part of $\vec{r}$, and the y-part of $\vec{v}$ only on the y-part of $\vec{r}$. They never interact.

$\Rightarrow$ The velocity vector always points tangent to the path, in the direction of motion.

$$\text{Speed} = |\vec{v}| = \sqrt{(v_x^2 + v_y^2)}$$

Example: -

Q) $\vec{r}(t) = 3t\hat{i} + (4t - t^2)\hat{j}$. Find $\vec{v}$ and $\vec{a}$.

Sol:

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \frac{d(3t)}{dt}\hat{i} + \frac{d(4t - t^2)}{dt}\hat{j}$$

$$\vec{v}(t) = 3\hat{i} + (4 - 2t)\hat{j}$$

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d(3)}{dt}\hat{i} + \frac{d(4 - 2t)}{dt}\hat{j}$$

$$\vec{a} = 0\hat{i} + (0-2)\hat{j}$$

$$\vec{a} = -2\hat{j}$$

$$\vec{v}(t) = 3\hat{i} + (4-2t)\hat{j}$$

$$\vec{a}(t) = -2\hat{j}$$